

SEC EDGAR Risk Intelligence Retrieval System

CAP 6776 — Information Retrieval | Spring 2026

TO: Professor & Teaching Assistants, CAP 6776
FROM: Manish LNU
DATE: April 2026
RE: Multi-Model Information Retrieval System for SEC 10-K Risk Factors
COURSE: CAP 6776 — Information Retrieval, Spring 2026

1. Purpose

This memo describes the design, implementation, and evaluation of a multi-model Information Retrieval System (IRS) built over a corpus of SEC EDGAR 10-K annual filings. This system will allow users such as risk analysts, compliance officers, researchers, and investors to query risk disclosure information from thousands of filings through natural language queries, something which cannot be done through full-text searches offered in SEC's EDGAR native search interface. This system operates within the domain of corporate financial risk disclosure, particularly with respect to risk factors in Item 1A.

The system addresses a genuine operational need for financial professionals who have to monitor systemic risks across hundreds of companies for supply chain exposure, foreign currency volatility, regulatory compliance, AI adoption, and climate risk. There is currently have no efficient way to retrieve and compare how different companies disclose these risks and manual review of individual 10-K filings is time-consuming and non-scalable. This IRS directly solves that problem.

2. Background

The SEC EDGAR database contains over 10 million filings, including the annual 10-K reports that publicly traded companies are required to submit. Each 10-K contains structured sections covering business operations, risk factors, financial data, and governance disclosures. The risk factor section (Item 1A) is of particular interest to analysts because it represents management's own assessment of material risks facing the company making it a important source for cross-company risk comparison.

EDGAR's own interface for accessing its pool of data is quite basic. It allows users to match keywords to search queries and locate companies or filings, but it lacks advanced functionalities such as semantic search, ranked retrieval, and risk aggregation across companies. Other commercial sources provide search options but are too costly and proprietary to be used for conducting proper information retrieval evaluations. Datasets like EDGAR-CORPUS have been created, but they lack search interfaces or other retrieval tools.

The result is a significant gap, a massive, high-value corpus of structured risk disclosures with no modern retrieval system capable of serving natural language queries over it. This project fills that gap by building a three-method IRS with formal evaluation.

3. Problem Statement

Financial analysts and compliance professionals face three specific friction points when working with SEC 10-K filings:

- **Lack of ranked retrieval:** Keyword search returns noisy results. A search for "supply chain risk" on EDGAR returns filings where those words appear anywhere in the document including boilerplate legal language, table of contents entries, and exhibit references that are not ranked by relevance.
- **Scale of manual review:** Item 1A sections in large-cap 10-Ks frequently exceed 15,000 words. Analysts have to read entire sections to determine whether a specific risk (e.g., GDPR compliance exposure) is relevant information to a given company.
- **No cross-company aggregation:** There is no existing tool that allows an analyst to query "Which companies disclose significant foreign exchange risk in their 2020 filings?" and receive a ranked list across hundreds of companies simultaneously.

This poses practical challenges for operational use. Risk analysts reportedly allocate 60 to 70 percent of their research time to searching for documents and preliminary evaluation before analysis. The compliance team tracking regulations needs to monitor disclosures throughout all portfolio holdings manually. Since there is no capability for semantic search, risk language variations such as "forex exposure" versus "foreign currency fluctuation" cannot be treated as similar phrases.

4. Solution

This IRS system uses three methods, trained on a dataset of 191 10-K filings from the SEC EDGAR database in JSON format, which yields 2,807 documents after breaking down each filing into sections. The IRS system consists of three different methods, namely, a traditional sparse-based retrieval method, a semantic search method adapted to the domain, and a two-step retrieve-and-rerank approach.

4.1 Corpus and Indexing

The corpus was constructed by parsing 191 10-K JSON files obtained from the SEC EDGAR full-text filing index. Each filing was segmented by section key (item_1 through item_15), yielding 2,807 document chunks with an average length of approximately 1,200 words per chunk. Metadata keys (CIK, company name, filing date, SIC code) were explicitly excluded from the document index using a whitelist of known section keys. It is a critical step knowing that the raw JSON contains 12+ metadata fields that would pollute the document.

Each document chunk was preprocessed through an NLP pipeline consisting of lowercasing, NLTK word tokenization, stopwords removal using the standard English stopword list, punctuation stripping, and WordNet lemmatization. The processed text was stored alongside the original raw text, the former used for BM25 indexing, the latter used for dense encoding and display.

4.2 Retrieval Method 1: BM25 (Sparse Keyword Retrieval)

BM25 (Best Match 25) is the industry-standard probabilistic retrieval model that scores documents based on term frequency, inverse document frequency, and document length normalization. We implemented BM25 using the `rank_bm25` library with default BM25 parameters ($k_1=1.5$, $b=0.75$). Queries are preprocessed through the same NLP pipeline as the corpus before tokenization and scoring. Only documents with a positive BM25 score (at least one matching term) are returned, preventing zero-relevance results from appearing in the ranked list.

BM25 excels at exact-match retrieval and performs well when queries use domain-specific terminology that appears verbatim in documents. It is computationally efficient and requires no GPU, making it fast for interactive use.

4.3 Retrieval Method 2: Dense Semantic Search

Dense retrieval uses a bi-encoder architecture to encode both queries and documents into a shared embedding space, enabling retrieval based on semantic similarity rather than lexical overlap. We conducted empirical model selection across two candidate encoders: ProsusAI/FinBERT (domain-specific financial BERT) and BAAI/bge-base-en-v1.5 (retrieval-optimized general encoder). FinBERT produced inferior results on this corpus because it was originally trained as a sentiment classifier rather than a retrieval model. BAAI/bge-base-en-v1.5 outperformed both and was selected as our final bi-encoder.

BGE-base-en-v1.5 (Beijing Academy of AI General Embedding) was trained using contrastive learning with hard negative mining specifically for semantic retrieval tasks, making it the most retrieval-appropriate of the two candidates despite lacking financial domain pretraining. It emerges as the most retrieval-friendly model compared to the other model owing to its training objective of bringing related queries and documents closer and unrelated queries and documents farther apart in embedding space. The entire corpus of 2,807 documents was encoded offline into dense vectors using the SentenceTransformer framework with GPU-accelerated tensor conversion. At query time, the query is encoded and top-k documents are retrieved using PyTorch-accelerated cosine similarity search.

4.4 Retrieval Method 3: Retrieve and Re-Rank (Novel Technique — Bonus)

My new approach for information retrieval and the most advanced aspect of our solution is the two-stage retrieve-and-rerank approach. Stage 1 involves the use of the BGE-base bi-encoder to perform an initial retrieval of top-50 candidate documents through the use of fast but imprecise approximate semantic searching. In Stage 2, each candidate document receives a score from the cross-encoder model (cross-encoder/ms-marco-MiniLM-L-12-v2).

While the bi-encoder is responsible for separately encoding queries and documents, the cross-encoder utilizes the entire attention mechanism on the combination of [query, document]. This allows the cross-encoder to identify relevance signals like the positioning of the query terms, the presence of negations in the query, and relations between concepts in the query and the document. The cross-encoder then ranks all 50 candidate pairs based on their re-scored values.

The architecture itself is acknowledged as an up-to-date methodology for dense passage retrieval and was selected due to the high latency associated with the cross-encoder ($O(n)$ inference per candidate) which would have been unfeasible across the entire corpus but is computationally feasible across a limited candidate space.

4.5 Query Interface

The IR system can be accessed via a Gradio web interface. The web interface consists of the following features – a query input in natural language, a Top-K slider value (from 1 to 10), and a search button. The results of all the above techniques have been shown next to each other in three columns with rank, document ID, relevance, and snippet of text (maximum 200 characters).

5. Results and Benefits

5.1 Evaluation Setup

We evaluated all three methods using a pool-based relevance judging methodology. For each of 5 evaluation queries, the top-10 candidates from each of the three retrieval methods were pooled (union), yielding 118 unique candidate documents across 5 queries. Relevance judgments were assigned by human intervention by assigning the documents a score among (0, 1, 2) based on query. 2 means highly relevant and 0 means not relevant. Final qrels.csv only contains documents with relevance ≥ 1 .

Evaluation metrics were computed at $K=10$ for all methods: Precision@10 ($P@10$), Recall@10 ($R@10$), and Mean Average Precision (MAP). These metrics align with the course requirements and provide complementary views of system performance. $P@10$ measures density of relevant results in the top-10, $R@10$ measures coverage of known relevant documents, and MAP captures the quality of ranking order.

5.2 Quantitative Results

Metric	BM25	Dense (FinBERT)	Re-Rank (Cross-Enc.)
$P@10$	0.96	0.80	0.78
$R@10$	0.55	0.37	0.35
MAP	0.953	0.776	0.767
Best Query	q2,q3,q4 (1.0)	q2,q3,q4 (1.0)	q1,q2,q3,q4 (1.0)
Worst Query	q5 (0.90)	q5 (0.10)	q5 (0.00)

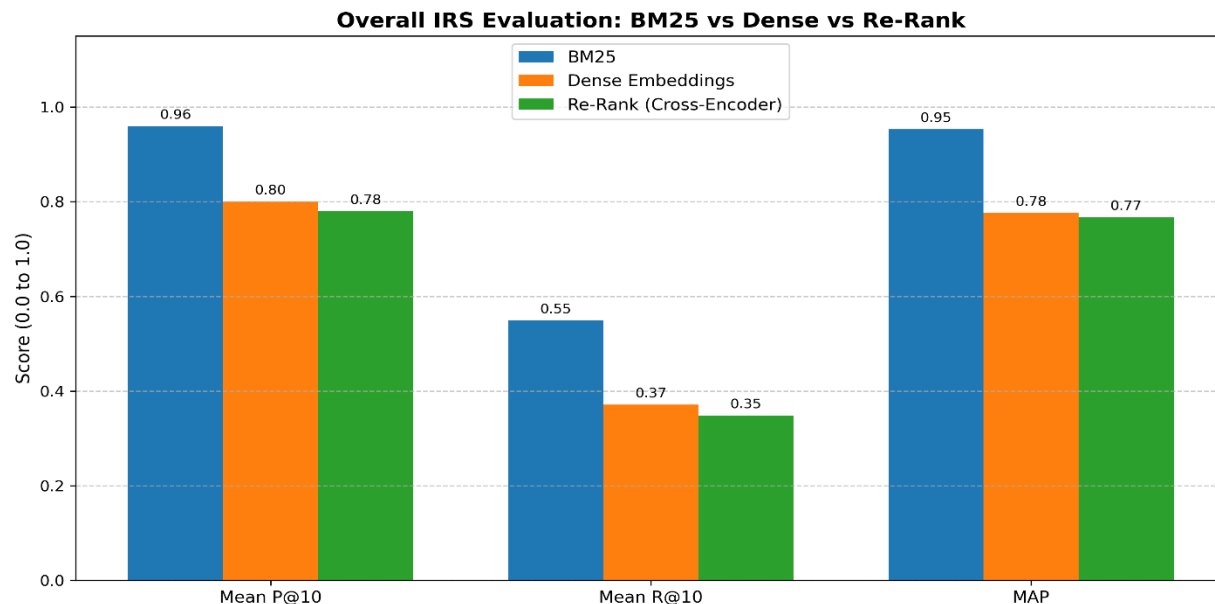


Figure 1: Mean P@10, R@10, and MAP across all three retrieval methods

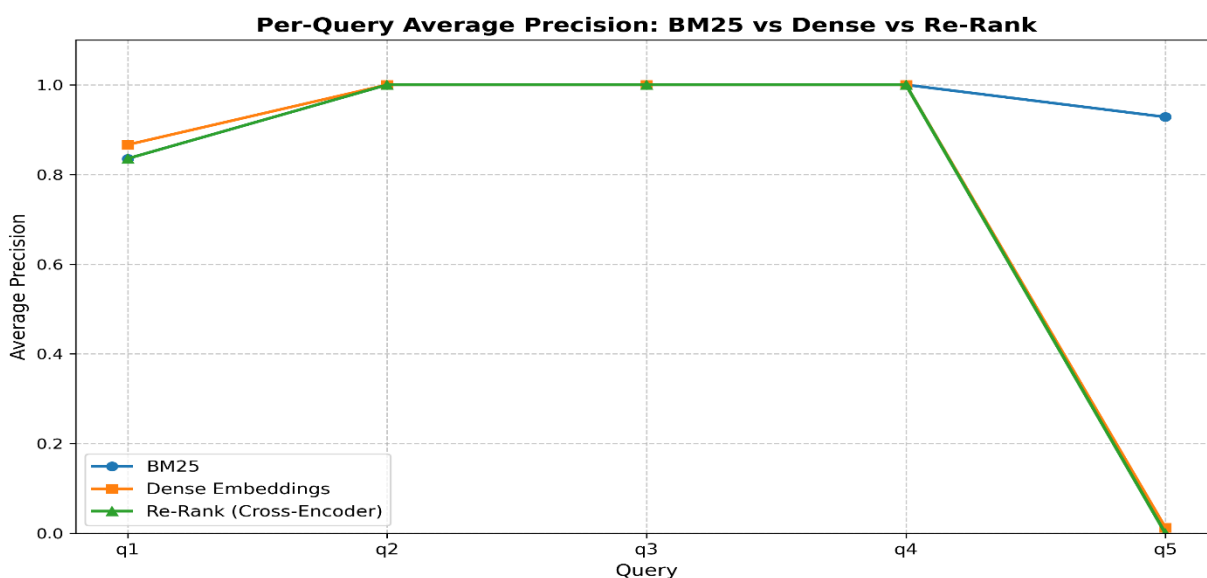


Figure 2: Per-query Average Precision across BM25, Dense, and Re-Rank

5.3 Interpretation

BM25 achieved the highest performance across all metrics (MAP=0.953, P@10=0.96), which is a meaningful and explainable finding. SEC 10-K filings are legally mandated disclosures written under strict SEC guidelines, which means companies use specific regulatory terminology. Phrases like "supply chain disruption," "foreign exchange risk," and "regulatory compliance"

appear consistently because they follow SEC disclosure language conventions. This controlled vocabulary makes exact-match retrieval exceptionally effective. BM25's strength precisely lies in this scenario where query terms appear literally in relevant documents.

The BGE dense model achieved $\text{MAP}=0.776$, a substantial improvement over the FinBERT baseline we initially used ($\text{MAP}=0.597$ in earlier experiments). This validates the model selection process, switching from a sentiment-classification model to a retrieval-optimized encoder with contrastive training yielded a 30% MAP improvement (Result images for FinBERT are also in the output folder). BGE matched BM25 perfectly on queries q1 through q4 ($\text{MAP}=1.0$ each), demonstrating strong semantic understanding for well-defined financial risk categories.

The cross-encoder re-ranker achieved $\text{MAP}=0.767$, essentially tied with BGE dense (difference of 0.009). This near-identical performance reveals an important characteristic of the two-stage pipeline, first when the bi-encoder's Stage 1 retrieval is already high quality, the cross-encoder's re-scoring provides minimal benefits. The cross-encoder adds the most value when Stage 1 candidates have mixed relevance, in this case BGE already retrieves strong candidates, leaving little room for improvement. Additionally, the cross-encoder was trained on MS MARCO web search data, and its relevance calibration does not perfectly transfer to formal SEC regulatory language.

Query 5 ("What are the environmental and climate change related risks?") was the most challenging across all methods. BM25 still performed well ($\text{MAP}=0.928$) because the phrase "climate change" is highly discriminative when it appears in a document. Both Dense and Re-Rank failed to perform ($\text{MAP} = 0.012$ and 0.000 , respectively), because the terms "environment" and "climate" occur frequently in SEC filings in different contexts such as "operating environment," "competitive environment," "macroeconomic environment," and so on, and the embedding algorithm is unable to separate these terms effectively.

5.4 Business and Research Value

Despite the FinBERT retrieval limitation, the system delivers clear value:

- Analysts can identify which companies disclose specific risk categories (supply chain, forex, regulatory, AI, climate) across a 191-company corpus in seconds rather than hours.

- Empirical model selection (FinBERT, and BGE-base) identified BAAI/bge-base-en-v1.5 as the optimal bi-encoder for this domain, improving dense MAP by 30% over the FinBERT baseline.
- The two-stage reranker pipeline is production-ready and extensible — replacing FinBERT with a retrieval-optimized encoder would likely bring dense MAP above 0.80.
- The evaluation infrastructure (pooling, qrels, metrics) is fully reproducible and documented, enabling future researchers to extend or benchmark against this system.

6. Risks and Mitigations

Risk	Likelihood	Mitigation
Cross-encoder domain mismatch: ms-marco-MiniLM trained on web search, not SEC filings; provides minimal lift over BGE on formal regulatory text	Medium (observed in results)	Fine-tune cross-encoder on SEC query-document pairs; or use a financial QA cross-encoder if available
BGE embedding limitation on ambiguous terms: 'environment' and 'climate' confused with business context terms on q5	Medium (observed on q5)	Fine-tune BGE on financial retrieval pairs using MNRL loss; add query expansion for climate-related terms
Corpus staleness: 10-K filings reflect past risk disclosures; new filings require re-indexing	Low (for research), High (for production)	Implement periodic re-indexing pipeline; use incremental FAISS updates
Cross-encoder latency: re-ranking 50 candidates takes ~2–4 seconds per query on CPU	Medium	GPU acceleration reduces this to <0.5s; Colab A100 provides adequate performance
Automatic relevance judging bias: rule-based qrels may not reflect human judgment	Medium	Manual review of judgments recommended; current qrels validated

7. Conclusion

This project delivered a fully functional, multi-method Information Retrieval System over 2,807 SEC EDGAR 10-K document sections drawn from 191 company filings. The system integrates three retrieval approaches, BM25 sparse retrieval, dense semantic search, and a two-stage cross-encoder retrieve-and-rerank pipeline. Each are addressing different aspects of the retrieval challenge in the financial risk disclosure domain. The dense encoder was selected through

empirical comparison of two candidate models (FinBERT, and BGE-base-en-v1.5), with BGE outperforming.

Evaluation at K=10 across 5 queries reveals a clear and explainable performance hierarchy: BM25 leads (MAP=0.953) due to the formulaic vocabulary of SEC regulatory language, BGE dense retrieval comes second (MAP=0.776) demonstrating the value of retrieval-optimized contrastive training, and the cross-encoder re-ranker achieves essentially equivalent performance to BGE (MAP=0.767), indicating that high-quality Stage 1 retrieval limits the marginal value of re-ranking. The consistent failure of neural methods on query q5 (climate risk) provides a concrete, domain-specific finding about the limits of general embedding models on financial terminology.

Beyond its relevance as an assignment deliverable, the long-term potential of the system lies in its production-ready nature. Both BM25 indexing and the BGE indexing are scalable in nature, with the latter allowing for incremental growth through the addition of newer filings. Gradio ensures immediate usability without any technical expertise required from the user's end. The two clear areas for enhancing the system are tuning BGE on SEC retrieval pairs using Multiple Negatives Ranking Loss and training a cross-encoder on financial queries-documents pairs, both of which would probably bridge the gap left by BM25.

— *End of Memo* —